DATA COMMUNICATIONS & NETWORKING: QUIZ 2

Topic: Digital Modulation (FSK, PSK, QAM, Constellations & Baud Rates)

Name: _____ Student ID: _____ Score: / 25

Question 1 (Calculation: FSK Half-Duplex)

An FSK signal operates at a bit rate of $N = 10$ kbps in half-duplex transmission mode. The two carrier frequencies are separated by $2\Delta f = f_{c1} - f_{c0} = 4$ kHz.

- Baud rate (S) = _____ KBaud
- Minimum transmission bandwidth ($BW = (f_{c1} - f_{c0}) + S$) = _____ kHz

Question 2 (Calculation: FSK Full-Duplex)

An FSK system operates in full-duplex transmission mode. The total medium bandwidth is 10 kHz, the upper carrier frequency is $f_{c1} = 20$ kHz, and the data rate is $N = 2$ kbps.

- Bandwidth allocated for each direction = _____ kHz
- Baud rate (S) = _____ kHz (or KBaud)
- Lower carrier frequency ($f_{c0} = S + f_{c1} - BW_{\text{dir}}$) = _____ kHz

Question 3 (Calculation: FSK Spectrum Parameters)

An FSK signal has an available bandwidth spanning from 1000 kHz to 3000 kHz. Carrier separation is $2\Delta f = f_{c1} - f_{c0} = 1000$ kHz. Assume $d = 1$ in $BW = (1 + d)S + 2\Delta f$.

- Total bandwidth (BW) = _____ kHz
- Midpoint of the bandwidth = _____ kHz
- Lower carrier frequency (f_{c0}) = _____ kHz
- Upper carrier frequency (f_{c1}) = _____ kHz
- Baud rate (S) = _____ KBaud
- Data rate (N) = _____ kbps

Question 4 (Bandwidth Comparison: Fixed Data Rate)

Determine the required bandwidth in Hertz (Hz) to transmit a data rate of $N = 5000$ bps with roll-off factor $d = 1$ for the following schemes:

- Binary ASK ($BW = (1 + d)S$): _____ Hz
- FSK with carrier separation $2\Delta f = 4$ kHz: _____ Hz
- QPSK (4-PSK, $r = 2$): _____ Hz
- 16-QAM ($r = 4$): _____ Hz

Question 5 (Calculation: Bits per Baud)

Using the relation $r = \log_2(L)$, determine the number of bits carried per baud (r) for:

- ASK with 8 different amplitude levels: _____ bits/baud
- FSK with 4 distinct signaling frequencies: _____ bits/baud
- PSK with 16 distinct phase angles: _____ bits/baud
- QAM with a constellation of 256 points: _____ bits/baud

Question 6 (Calculation: Data Rate from Baud Rate)

Given a fixed symbol transmission rate of $S = 2000\ \mathrm{baud/s}$, calculate the data rate N in kbps ($N = r \times S$):

- Binary FSK ($r = 1$): _____ kbps
- Binary ASK ($r = 1$): _____ kbps
- 4-ASK ($r = 2$): _____ kbps
- 16-QAM ($r = 4$): _____ kbps

Question 7 (Calculation: FSK with Filter Factor $d = 1$)

An FSK system transmits data at $N = 25\ \mathrm{kbps}$ with carrier separation $2\Delta f = 12\ \mathrm{kHz}$. If filter parameter $d = 1$, what is the required bandwidth?

- Baud rate (S) = _____ KBaud
- Required bandwidth ($BW = (1 + d)S + 2\Delta f$) = _____ kHz

Question 8 (Calculation: Higher-Order QAM)

A high-speed microwave link uses 64-QAM modulation. The channel symbol rate is $S = 12\ \mathrm{MBaud}$.

- Bits per symbol ($r = \log_2(64)$) = _____ bits
- Bit rate (N) = _____ Mbps

Question 9 (Constellation Diagram Analysis)

A constellation diagram consists of 16 constellation points arranged on two concentric circles: 8 points on the inner circle and 8 points on the outer circle.

- Number of amplitude levels used = _____
- Number of phase shifts used = _____
- Bits carried per symbol (r) = _____ bits/symbol

Question 10 (Spectral Efficiency Comparison)

We must transmit a digital video feed at $N = 24\ \text{Mbps}$ across a band-pass channel with filter roll-off factor $d = 0$. Compare the required bandwidth for:

- BPSK ($r = 1$): $BW =$ _____ MHz
- QPSK ($r = 2$): $BW =$ _____ MHz
- 16-QAM ($r = 4$): $BW =$ _____ MHz
- 64-QAM ($r = 6$): $BW =$ _____ MHz

Question 11 (Multiple Choice: Modulation Characteristics)

Why is Quadrature Amplitude Modulation (QAM) predominantly favored over high-order Phase Shift Keying (such as 64-PSK) for high-speed data transmission?

[A] High-order PSK requires higher carrier frequency than QAM

[B] Arranging constellation points across both amplitude and phase creates greater geometric distance between adjacent points, lowering susceptibility to phase noise

[C] QAM transmitters do not require analog digital-to-analog converters

[D] PSK signals are completely immune to noise whereas QAM is strictly analog

Question 12 (Comprehensive Problem: Voice-Band Modem Design)

A telephone line channel provides an effective passband of $BW = 3000\ \text{Hz}$. Filter factor is $d = 0.2$.

- Maximum baud rate ($S = \frac{BW}{1+d}$) = _____ Baud
- Achievable bit rate if using 16-QAM = _____ bps
- Achievable bit rate if using 256-QAM = _____ bps

Answer Key & Detailed Explanations

Question 1: $S = 10\ \text{KBaud}$; $BW = 14\ \text{kHz}$
Mathematical Derivation: In binary FSK half-duplex, $r = 1 \implies S = N = 10\ \text{KBaud}$.

$$BW = (f_{c1} - f_{c0}) + S = 4\ \text{kHz} + 10\ \text{KBaud} = 14\ \text{kHz}$$

[cite: 1]

Question 2: **Bandwidth/direction =** $5\ \text{kHz}$; $S = 2\ \text{kHz}$ **(**$2\ \text{KBaud}$**);** $f_{c0} = 17\ \text{kHz}$
Mathematical Derivation:

$$BW_{\text{dir}} = \frac{10\ \text{kHz}}{2} = 5\ \text{kHz}$$

Baud rate equals bit rate: $S = N = 2\ \text{kbps} = 2\ \text{KBaud}$.

$$BW_{\text{dir}} = S + (f_{c1} - f_{c0}) \implies f_{c0} = S + f_{c1} - BW_{\text{dir}} = 2\ \text{kHz} + 20\ \text{kHz} - 5\ \text{kHz} = 17\ \text{kHz}$$

[cite: 1]

Question 3: $BW = 2000\ \text{kHz}$; **Midpoint =** $2000\ \text{kHz}$; $f_{c0} = 1500\ \text{kHz}$; $f_{c1} = 2500\ \text{kHz}$; $S = 500\ \text{KBaud}$; $N = 500\ \text{kbps}$
Mathematical Derivation:

$$BW = 3000\ \text{kHz} - 1000\ \text{kHz} = 2000\ \text{kHz}$$

$$\text{Midpoint} = \frac{1000 + 3000}{2} = 2000\ \text{kHz}$$

Given $2\Delta f = 1000\ \text{kHz}$ symmetric about the midpoint:

$$f_{c0} = 2000 - \frac{1000}{2} = 1500\ \text{kHz}, \quad f_{c1} = 2000 + \frac{1000}{2} = 2500\ \text{kHz}$$

Solving for baud rate with $d = 1$:

$$BW = (1 + d)S + 2\Delta f \implies 2000 = (1 + 1)S + 1000 \implies 2S = 1000 \implies S = 500\ \text{KBaud}$$

Since binary FSK carries $1\ \text{bit/baud}$:

$$N = S = 500\ \text{kbps}$$

[cite: 2]

Question 4: ASK $= 10000$ Hz; FSK $= 14000$ Hz; QPSK $= 5000$ Hz; 16-QAM $= 2500$ Hz
Mathematical Derivation: Given $N = 5000$ bps, $d = 1$:

$$\text{ASK: } S = N = 5000 \text{ Baud} \implies BW = (1+d)S = 2 \times 5000 = 10,000 \text{ Hz}$$

$$\text{FSK: } BW = (1+d)S + 2\Delta f = (2 \times 5000) + 4000 = 14,000 \text{ Hz}$$

$$\text{QPSK: } r = \log_2(4) = 2 \implies S = \frac{5000}{2} = 2500 \text{ Baud} \implies BW = (1+d)S = 2 \times 2500 = 5000 \text{ Hz}$$

$$\text{16-QAM: } r = \log_2(16) = 4 \implies S = \frac{5000}{4} = 1250 \text{ Baud} \implies BW = (1+d)S = 2 \times 1250 = 2500 \text{ Hz}$$

[cite: 3]

Question 5: 8-ASK $= 3$ bits; 4-FSK $= 2$ bits; 16-PSK $= 4$ bits; 256-QAM $= 8$ bits
Mathematical Derivation:

$$r = \log_2(L)$$

$$\log_2(8) = 3 \text{ bits}, \quad \log_2(4) = 2 \text{ bits}, \quad \log_2(16) = 4 \text{ bits}, \quad \log_2(256) = 8 \text{ bits}$$

[cite: 3]

Question 6: FSK $= 2$ kbps; ASK $= 2$ kbps; 4-ASK $= 4$ kbps; 16-QAM $= 8$ kbps
Mathematical Derivation:

$$N = r \times S$$

$$\text{FSK: } N = 1 \times 2000 = 2000 \text{ bps} = 2 \text{ kbps}$$

$$\text{ASK: } N = 1 \times 2000 = 2000 \text{ bps} = 2 \text{ kbps}$$

$$\text{4-ASK: } r = \log_2(4) = 2 \implies N = 2 \times 2000 = 4000 \text{ bps} = 4 \text{ kbps}$$

$$\text{16-QAM: } r = \log_2(16) = 4 \implies N = 4 \times 2000 = 8000 \text{ bps} = 8 \text{ kbps}$$

[cite: 4]

Question 7: $S = 25$ KBaud; $BW = 62$ kHz
Mathematical Derivation:

$$S = N = 25 \text{ KBaud}$$

$$BW = (1+d)S + 2\Delta f = (1+1) \times 25 \text{ kHz} + 12 \text{ kHz} = 50 + 12 = 62 \text{ kHz}$$

Question 8: $r = 6$ bits/symbol; $N = 72$ Mbps
Mathematical Derivation:

$$r = \log_2(64) = 6 \text{ bits/symbol}$$

$$N = r \times S = 6 \times 12 \text{ MBaud} = 72 \text{ Mbps}$$

Question 9: **Amplitude levels = 2; Phase shifts = 8; Bits/symbol = 4**
Mathematical Derivation: Two concentric rings represent 2 distinct carrier amplitudes. Each ring contains 8 points, meaning 8 distinct phase angles are used.

$$L = 16 \implies r = \log_2(16) = 4 \text{ bits/symbol}$$

Question 10: $\text{BPSK} = 24$ MHz; $\text{QPSK} = 12$ MHz; $\text{16-QAM} = 6$ MHz; $\text{64-QAM} = 4$ MHz
Mathematical Derivation: With $d = 0$, $BW = S = \frac{N}{r}$:

$$\text{BPSK } (r = 1) : BW = \frac{24}{1} = 24 \text{ MHz}$$

$$\text{QPSK } (r = 2) : BW = \frac{24}{2} = 12 \text{ MHz}$$

$$\text{16-QAM } (r = 4) : BW = \frac{24}{4} = 6 \text{ MHz}$$

$$\text{64-QAM } (r = 6) : BW = \frac{24}{6} = 4 \text{ MHz}$$

Question 11: **[B] Arranging constellation points across both amplitude and phase creates greater geometric distance between adjacent points, lowering susceptibility to phase noise**
Explanation: In M-PSK, all points lie crowded on a single circle. As M grows, the angular distance between neighboring points shrinks rapidly ($\Delta\theta = 360^\circ/M$), making it vulnerable to phase jitter. QAM spans a 2D grid, maximizing Euclidean distance between points for the same average power.

Question 12: $S = 2500$ Baud; $\text{16-QAM} = 10,000$ bps; $\text{256-QAM} = 20,000$ bps
Mathematical Derivation:

$$BW = (1 + d)S \implies S = \frac{BW}{1 + d} = \frac{3000}{1 + 0.2} = \frac{3000}{1.2} = 2500 \text{ Baud}$$

$$\text{16-QAM } (r = 4) : N = 4 \times 2500 \text{ Baud} = 10,000 \text{ bps} = 10 \text{ kbps}$$

$$\text{256-QAM } (r = 8) : N = 8 \times 2500 \text{ Baud} = 20,000 \text{ bps} = 20 \text{ kbps}$$
